

# VoiceForge Master Studio (v1.3.4)

## Official User Manual & Operational Runbook

---

### Table of Contents

1. Product Overview & Architecture
  2. Free vs. PRO Feature Matrix
  3. Windows 11 Setup & First-Time Launch
  4. Master Studio Dashboard & Controls
  5. AI Audio Engines & Model Manager
  6. Voice Vault & Custom .vfs Audio Cloning
  7. Live Chat Ingest & Real-Time Commands
  8. Speech Moderation, Anti-Spam & Whitelisting
  9. OBS Studio Chroma-Key HUD Integration
  10. Troubleshooting & Diagnostics
- 

## 1. Product Overview & Architecture

**VoiceForge Master Studio** is a real-time, hybrid AI Text-to-Speech (TTS) livestream broadcasting suite. It bridges live stream chat feeds from multiple platforms (Rumble, Kick, X.com, Twitch, YouTube) into high-fidelity neural audio.

### Core Systems

- **Native C++ Supervisor (VoiceForgeStudio.exe):** A high-priority Windows process wrapper that verifies hardware capability, isolates dependencies, manages single-instance sockets, and uses Windows Job Objects to prevent orphaned background processes.
  - **Dual-Engine Audio Dispatcher:**
    - **Kokoro-82M:** Lightweight base neural engine (54 multilingual voices).
    - **Custom Neural Cloning Engine:** Heavyweight zero-shot flow-matching models (Chatterbox-Turbo, CosyVoice 2, Qwen3-TTS) for cloning custom voices from reference audio.
  - **Multicast Live Chat Ingest:** Headless browser automation that scrapes chats and intercepts raw WebSocket frames with video stream suppression to save GPU/CPU cycles.
  - **OBS Chroma-Key HUD (reader.html):** Real-time broadcast overlay with customizable chroma-key backgrounds and platform badges.
- 

## 2. Free vs. PRO Feature Matrix

VoiceForge Studio operates on a two-tier licensing model: **Free** (unlocked by default) and **PRO** (unlocked via Gumroad license key).

| Feature                                       | Free Tier                                                                          | PRO Tier (Gumroad)                                                                  |
|-----------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Kokoro-82M Neural Engine                      | Full Access (54 Voices)                                                            | Full Access (54 Voices)                                                             |
| Multilingual Language Packs                   |  |  |
| Real-Time Speed Tags ( - .75 , - 1.20 )       | Supported                                                                          | Supported                                                                           |
| Multicast Chat Ingest (Rumble, Kick, X, etc.) | Supported                                                                          | Supported                                                                           |
| OBS Chroma-Key Chat HUD                       | Supported                                                                          | Supported                                                                           |
| Speech Moderation & Spam Shield               | Supported                                                                          | Supported                                                                           |
| Chatterbox-Turbo Cloning Engine               | Locked (Routes to Kokoro)                                                          | Unlocked                                                                            |
| CosyVoice 2 (0.5B) Flow-Matching              | Locked                                                                             | Unlocked                                                                            |
| Qwen3-TTS Multi-Emotion Engine                | Locked                                                                             | Unlocked                                                                            |
| Custom Cloned Voices (voices/ folder)         | Read with Kokoro default                                                           | Full Neural Cloning                                                                 |
| Proprietary .vfs Audio Package Support        | Read with Kokoro default                                                           | Full Neural Cloning                                                                 |
| Per-Voice Personality Studio                  | Locked                                                                             | Unlocked                                                                            |

How to Activate PRO

- 1. Click the amber ★ Upgrade to PRO button in the top navigation header.
- 2. Enter your Gumroad purchase license key.  
(For testing/developer use, enter the master developer key: VF-PRO-2026).
- 3. Click ★ Activate PRO.
- 4. The button turns into a green ★ PRO ACTIVE badge, and all custom cloning engines unlock instantly.

3. Windows 11 Setup & First-Time Launch

Step 1: Unblock the Zip Archive

Because the application is downloaded via a web browser, Windows tags it with **Mark of the Web (Zone.Identifier)**:

- 1. Right-click VoiceForgeStudio-Windows-x64.zip

→

**Properties.**

- At the bottom, check the **"Unblock"** checkbox

→

Click **Apply / OK**.

- Extract the `.zip` archive to your preferred directory (e.g., `C:\VoiceForgeStudio\`).

## Step 2: Automatic First-Time Setup

Double-click **VoiceForgeStudio.exe** (or **run\_voiceforge.bat**):

- A dialog appears: *"Welcome to VoiceForge Master Studio! First-time setup is required to download the portable runtime."*
- Click **OK**.
- A Command Prompt window opens, utilizing Windows' native `curl.exe` and `tar.exe` to provision:
  - Embedded Python 3.11
  - PyTorch (CUDA 12.1 GPU acceleration)
  - Base audio libraries & Kokoro-82M
- Once completed, the setup window closes automatically, the server starts, and VoiceForge opens in its own **Dedicated Desktop Window**.

## 4. Master Studio Dashboard & Controls

code

```
+-----+
+-----+
| [icon] VoiceForge  [ Stop Reading / Start Reading]  (● LIVE)  [🔊 Audio
Active]  [🔴 Exit] |
| [LIVE LOG]: [Chatterbox Clone] Synthesizing 'apple' using reference: Apple.vfs
|
+-----+
+-----+
| [🎤 TTS Prompter] [📖 Reader] [⌚ Live Activity] [📝 Audio Log] [📡 Sources] [⚙️
Settings]      |
+-----+
+-----+
```

### Top Navigation Bar

- Master Stream Switch:** Toggle live reading on/off. Shortcut: `Ctrl + Space`.

- **Audio Unlocker Button:** Browsers require a user gesture before playing audio. If the button reads  Click to Enable Audio, click it once to turn it green ( Audio Active).
-  **Exit Studio:** Safely terminates the Python server, shuts down Playwright browser instances, frees port 8080, and closes the window cleanly.

## Live Activity Ticker & Terminal Log

Positioned directly beneath the header, the dark console bar displays live server events:

- **Green pulse dot:** Server healthy and connected.
- **Status ticker:** Shows active voice synthesis, chat extraction, or download progress.
-  **Open Terminal Log:** Opens a live scrolling console modal inside the interface.

## 5. AI Audio Engines & Model Manager

Navigate to **Settings**

→

 **Model Weights & Disk Footprint Manager.**

### Dual-Engine Switching Logic

VoiceForge uses a hybrid dispatcher:

- **Kokoro-82M:** Always active as the primary multilingual base engine.
- **Custom Models:** Only **one** large cloning engine is active at a time to manage VRAM usage.

code

|                                                                                                                                                                                               |                                                                                                                                                                                               |                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [ Kokoro-82M ]                                                                                                                                                                                | [ Chatterbox-Turbo ]                                                                                                                                                                          | [ CosyVoice 2 (0.5B) ]                                                                                                                                                                            |
|  Always Active Base                                                                                        |  Active Custom Engine                                                                                      |  Activate This Engine                                                                                        |
| ✓ Installed (312.6 MB)                                                                                                                                                                        | ✓ Installed (1200.0 MB)                                                                                                                                                                       | ✓ Installed (3800.0 MB)                                                                                                                                                                           |
| [  Folder ] [  Remove ] | [  Folder ] [  Remove ] | [  Folder ] [  Remove ] |
| ]                                                                                                                                                                                             |                                                                                                                                                                                               |                                                                                                                                                                                                   |

## Managing Models

### 1. Downloading Models:

- Click  **Auto-Download** on any model card.
- The top live log bar shows the real-time progress:  
[Downloading] CosyVoice 2 (0.5B): 42% (1600.0 MB / ~3800 MB)
- Upon completion, the card updates to show an activation button.

### 2. Switching Engines:

- Click  **Activate This Engine** on an installed engine.

- That engine turns green ( **Active Custom Engine**), and the previous engine switches to red.
- All custom cloned voices (, , etc.) will route through the newly active engine.

### 3. Manual / Offline Installation (Restricted Networks):

- If a machine has firewall restrictions, click  **Offline / Manual** on any model card.
- Copy the target folder path shown in the popup (e.g., `pretrained_models\chatterbox-turbo\`).
- Download the model files via browser using the direct link provided.
- Paste the files into the folder and click  **I Placed the Files – Verify Now**. The app validates the files immediately.

### 4. Deleting Models to Reclaim Disk Space:

- Click  **Remove** on any inactive model.
- VoiceForge uses a force-delete process that strips Windows read-only locks, deleting the files from both `pretrained_models/` and `~/.cache/huggingface/hub/`.

## 6. Voice Vault & Custom .vfs Audio Cloning

### Supported Audio Formats

Custom voices placed in the `voices/` directory are recognized in the following formats:

- **.vfs (VoiceForge Studio Track)**: Disguised MP3/WAV audio track.
- **.mp3**: Standard MP3 audio.
- **.wav**: Uncompressed PCM audio.
- **.flac / .ogg**: Lossless and Ogg audio.

### Adding a New Voice

1. Record a **5 to 15-second** reference audio sample of a speaker talking in a clear tone with minimal background noise.
2. Save the file with the voice name (e.g., `ron.vfs`, `Apple.vfs`, or `swifty.mp3`).
3. Place the file inside the `voices/` folder.
4. Open the **Voice Studio** tab and click  **Refresh**. The voice is loaded into memory.

### Previewing Voices (Verbal Model Announcements)

Click ► **Preview** next to any voice in the Voice Studio:

- The button updates to  **Loading...** while the neural network processes the audio.
- The voice announces which engine generated it:
  - *"Hello! This is Bane, running live on Chatterbox Turbo."*
  - *"Hello! This is Bella, running live on Kokoro 82M."*

## 7. Live Chat Ingest & Real-Time Commands

Navigate to the  **Sources** tab to connect stream chats.

### Connecting a Platform

1. Click **+ Add Source**.
2. Select your platform from the dropdown:
  - **Rumble**: Paste the Live Stream URL or Chat Popup URL (`https://rumble.com/chat/popup/...`).
  - **Kick**: Paste the channel URL or username (`https://kick.com/username`).
  - **X.com (Twitter)**: Paste the live broadcast or profile URL.
  - **Twitch**: Paste the Twitch channel URL.
  - **YouTube Live**: Paste the YouTube live stream URL.
3. Click **Connect Source**. The status will display **CONNECTING**

→  
**CONNECTED**.

### Ingest Rules & Fallbacks

- **Un-tagged chat messages** are spoken using your **Default Voice** (configured in the Sources tab).
- **Viewers calling a voice** use the `!name` prefix (e.g., `!bane Hello everyone`).
- **Speed modifiers** can be appended to voice tags:
  - `!bane-.75 Slow speech` (Speaks at **0.75x speed**).
  - `!bella-1.20 Fast speech` (Speaks at **1.20x speed**).
  - `!swifty-0.80 Normal speed` (Speaks at **0.80x speed**).
- **Multi-voice sentences** are split and synthesized consecutively:
  - `!bane-.75 I am slow. !bella-1.20 And I am speaking fast!`

## 8. Speech Moderation, Anti-Spam & Whitelisting

Navigate to the  **Audio Log** tab to manage chat moderation rules.

code

|                              |                                                        |
|------------------------------|--------------------------------------------------------|
| [X] Bot Shield (Block Links) | [X] Keyword Censorship (Spoken Text <b>Only</b> )      |
| BLOCKED MESSAGE KEYWORDS:    | spam, giveaway, <b>free</b> followers, discord.gg/scam |
| WHITELISTED USERS:           | (Clean <b>by default</b> - <b>add</b> usernames here)  |
| MANUALLY BANNED USERS:       | (Clean <b>by default</b> - <b>add</b> usernames here)  |

### Moderation Protections

1. **Spoken-Text-Only Filtering:** Keywords only filter the **message text**. A chatter whose username happens to contain a keyword will never be blocked.
2. **Whitelist Priority:** Users listed in the Whitelist bypass all bot, length, and keyword rules.
3. **Spam Shield (8-Second Duplicate Detection):**
  - If a user posts the exact same message twice within **8 seconds**, the duplicate is blocked from audio.
  - Instead of being discarded silently, the blocked duplicate appears in the **Filtered / Blocked Messages Log** table.
  - You can click  **Restore** next to any message to speak it immediately.

## 9. OBS Studio Chroma-Key HUD Integration

VoiceForge includes a transparent, real-time chat overlay designed for OBS Studio.

code

```
+-----+
| [19:25:01] [RUMBLE] TheeSleepybear: !Reeves scrimp is on sale |
| [19:25:08] [RUMBLE] The02Tones: MissBear disputes |
| [19:26:00] [KICK] SaltyGamer: what model is that? |
+-----+
```

### Adding to OBS Studio

1. In OBS Studio, add a new **Browser Source**.
2. Set the URL to: **http://localhost:8080/reader**
3. Set **Width:** 600 (or 800), **Height:** 900.
4. Right-click the Browser Source
 

→

**Filters** → Add **Chroma Key**:
  - **Key Color Type:** Green (#00FF00).
  - The green background disappears, leaving the animated chat lines and platform badges on screen.
5. In the toolbar at the top of /reader, you can adjust the font family, font size, text color, and background key color.

## 10. Troubleshooting & Diagnostics

### Issue 1: Server fails with [Errno 10048] Address already in use

- **Cause:** An earlier instance of VoiceForgeStudio.exe or python.exe is still running in the background and holding port 8080.

- **Fix:**

1. Open Command Prompt on Windows and run:

```
taskkill /F /IM python.exe /IM VoiceForgeStudio.exe
```

2. Launching via `run_voiceforge.bat` automatically runs this command before boot to prevent port conflicts.

---

## Issue 2: Smart App Control (SAC) blocks VoiceForgeStudio.exe

- **Cause:** Windows attached the "Mark of the Web" tag to the downloaded .zip file.

- **Fix:**

1. Open PowerShell inside your VoiceForge folder.
2. Run the unblock command:

```
Get-ChildItem -Recurse | Unblock-File
```

3. Alternatively, launch the app via `run_voiceforge.bat`, which runs directly through the local embedded Python engine and bypasses binary blocks.

---

## Issue 3: Chat messages are received, but no sound plays

- **Checklist:**

1. Look at the top navigation bar: does the button read  **Click to Enable Audio**? Browsers block audio until clicked. Click it once to activate sound.
2. Check the master stream switch: does it read  **Start Reading** (Paused)? Click it or press `Ctrl + Space` to switch to  **Stop Reading** (Live).
3. Check the volume slider: ensure the volume is not set to 0% or muted ().

---

## Issue 4: RuntimeError: expected scalar type Double but found Float

- **Cause:** An upstream bug in `chatterbox-tts` where `norm_loudness` promotes waveforms to 64-bit float tensors.
- **Fix:** This is patched in `engine_manager.py`. Ensure your repository has the latest commit (`patch_chatterbox_dtype.py`).

---

## Issue 5: A specific viewer's chats (e.g., BearLoves) are not being read



- **Checklist:**

1. Verify the **Whitelist** in the Audio Log tab. Check for spelling differences (e.g., bear`l`overs vs. bear`r`loves).
  2. The updated Rumble DOM parser extracts chatters whether they are channel owners, subscribers, or non-linked guests. Check the **Live Terminal Bar** at the top—when a message is read, it logs [Chat Ingest] [RUMBLE] Username: Message.
- 

## Issue 6: Custom voices play the Kokoro voice instead of cloning

- **Cause:** The custom voice engine has not been unlocked with PRO, or its model weights have not been downloaded.
- **Fix:**
  1. Check the header: does it show  **PRO ACTIVE**? If not, click  **Upgrade to PRO** and enter your key (or VF-PRO-2026).
  2. Open **Settings**  
  
→  
**Model Manager:** ensure **Chatterbox-Turbo** shows  **Active Custom Engine**. If it displays  **Auto-Download**, click it to download the weights.